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# AI-integrated interactive learning system for enhancing cybersecurity education

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Cybersecurity education often struggles to provide scalable, practice-oriented learning experiences while maintaining pedagogical oversight. This paper presents AI-CES, a lecturer-in-the-loop, AI-assisted interactive learning platform designed to support scenario-based cybersecurity education. The system combines AI-generated simulation drafting, lecturer review and approval, student-facing multi-round simulations, and performance tracking within a single web-based environment. We report the design and preliminary evaluation of the prototype rather than a validated effectiveness study. An exploratory evaluation was conducted with 30 participants from one computer science department, including 10 lecturers and 20 students, using role-specific post-use surveys with Likert-scale and open-ended items. Descriptive statistics and thematic coding were used to analyse perceived usability, engagement, content quality, and educational value. Participants generally viewed the platform positively, particularly its interactivity, immediacy of feedback, and lecturer oversight of AI-generated content. Lecturers also valued the visibility provided by the analytics dashboard. However, the study does not establish causal effects on learning outcomes, teaching effectiveness, or long-term skill development. Findings should, therefore, be interpreted as evidence of feasibility and user acceptability in a pilot setting. Limitations include the small single-site sample, reliance on self-reported perceptions, and dependence on third-party AI services. Future work should incorporate controlled comparisons, pre/post knowledge measures, multi-institution evaluation, and longitudinal analysis of learner interaction data.

KEYWORDS
**AI-driven Cybersecurity Education System (AI-CES), artificial intelligence (AI), cybersecurity students (CS-students), cybersecurity teacher (CS-Teachers), GenAI-driven intelligent tutoring systems (GenAI ITSs), generative artificial intelligence (GenAI), large language model (LLM), research questions (RQs)**

## 1 Introduction

Cybersecurity has become a crucial discipline, and the current traditional lecture-based teaching methods usually provide a minimal hands-on experience (Crabb et al., 2024). A recent review of U.S. cybersecurity programs highlights insufficient evaluation of instructional methods and urges educators to adopt more current tools and frameworks (Crabb et al., 2024). Consequently, cybersecurity programs may not consistently to provide adequate theoretical and practical training of cybersecurity concepts, leaving graduates without the experience and confidence necessary to perform effectively in real cybersecurity settings. At the same time, AI is changing learning and teaching practices. Many educators

now see that AI tools can improve learning experiences and should not be ignored as Hodges and Ocak stated that "using AI in higher education must not be ignored" (Hodges and Ocak, 2023). The usage of AI is particularly significant in cybersecurity education, offering immediate tutoring support to both teachers and students. For example, an AI system could generate some new network attack scenarios or even give some hints to students who are having difficulty with vulnerability analysis tasks. Park and Simmons (2024) suggested that incorporating AI capabilities allows instructors to keep these topics more up-to-date with a more tailored learning experience, even as new cyber threats emerge. Although AI-powered tools can greatly enhance cybersecurity education, their effectiveness still relies on human oversight. A foremost concern is ensuring the accuracy, transparency and security of AI-generated content, which can lead to misleading information (Park and Simmons, 2024). If not addressed, these inaccuracies may confuse learners and spread misconceptions (Gandolfi, 2025). Therefore, AI tools should serve under an instructor's full supervision, rather than replacing lecturers, and that is what this project aims for.

The main aim of this project is to enhance cybersecurity education, helping students personalize their learning experience and making their learning process more efficient. This approach is intended to support improved learning experiences for both students and lecturers by focusing on practical skills and the employability of cybersecurity graduates. The AI-Cybersecurity Enhancing System (AI-CES) provides students with a variety of practice exercises, such as how a red or blue team works, how an SQL-injection attack occurs or any other cybersecurity subjects, allowing them to apply concepts in realistic scenarios. Additionally, it adjusts the difficulty of tasks and offers hints based on individual performance, personalizing the learning experience to boost both competence and confidence. The system also includes performance tracking to identify areas where students are struggling. In a field where certifications and skills are crucial for employment, this approach offers continuous practice and feedback, helping students build a strong portfolio. Through this approach, the system aims to support the development of practical skills for graduates, making them better prepared for real-world challenges in the cybersecurity domain.

This study is guided by three primary RQs that shape the initial design and implementation of the AI-CES for the simulated cybersecurity topics. These questions focus on establishing a foundation for the content that can support rapid, scalable development while addressing educational and industry objectives: RQ1. How can a lecturer-in-the-loop GenAI architecture be designed to support the creation, approval, and delivery of scenario-based cybersecurity simulations? RQ2. How do lecturers and students perceive the usability, perceived content quality, and educational value of the AI-CES prototype? RQ3. What feasibility issues, limitations, and design implications emerge from the initial deployment of AI-CES? These questions focus on establishing the core aspects of the prototype of this system, which aims to explore a collaborative framework that addresses the skill requirements of cybersecurity professionals while enhancing the curriculum of cybersecurity programs.

The following objectives aim to strike a balance between industry relevance and educational value, creating a foundation for ongoing improvements within AI-CES: (i) Facilitating lecturers and students' collaboration: Design and create a prototype of an AI-based learning system called the Cybersecurity Education System (AI-CES) that helps students and teachers work together and allows them to interact with content generated by AI. (ii) Enhancing productivity and accessibility: Integrate OpenAI's model into the system to support the generation of customized cybersecurity learning materials under lecturer supervision. The system will include mechanisms for lecturers to review, approve, or modify AI-generated content before it reaches students, ensuring the material's accuracy and relevance. (iii) Supporting educational and industry alignment: Evaluate the educational experience for both students and lecturers by conducting a preliminary evaluation of the educational experience, assessing student perceptions and the potential of AI-CES to support learning outcomes. Additionally, examine how these systems enable teachers to focus more on engaging directly with their students rather than on repetitive content creation and answering basic questions. This assessment provides preliminary insights into how AI-CES may support student learning and teaching practices. These objectives support the AI-CES framework and promote its evolution through continuous collaboration among academics, program directors, industry partners, students, and graduates, supporting iterative refinement, and future investigation.

This paper is organized into five sections, beginning with an introduction that clarifies the primary concept behind this project. Section 2 identifies key features of similar tools and systems and related work that use GenAI to deliver personalized learning experiences, particularly in cybersecurity education. Section 3 describes the design and development approach of the AI-CES, highlighting its core requirements and methodology. Then, Section 4 highlights the key outcomes of the implementation phase in relation to the project's main objectives. Section 5 discusses the theoretical implications and the potential adaptability and applicability of the proposed AI-CES, addresses common challenges and short-term mitigation strategies, and outlines changes planned for future work.

# 2 Literature review

## 2.1 Cybersecurity education challenges

Recent studies highlight key challenges of cybersecurity education from the fast-evolving and complex nature of this field. A 2024 mixed-method study stated that curriculum must keep pace with ever-changing threats and technologies, yet, traditional teaching methods struggle to convey real-world skills in such a dynamic domain (Salem et al., 2024). Additionally, cybersecurity education requires interactive and adaptive teaching strategies to address low engagement and skills (Salem et al., 2024). The same study found that despite high interest in cybersecurity, students in an online program reported low engagement due to a lack of interactivity (Salem et al., 2024). This underlines that merely

delivering content is insufficient: active learning is needed to maintain enthusiasm. This field calls for interactive and adaptive teaching strategies to tackle low engagement, communication and skill gaps in cybersecurity education (Salem et al., 2024). Another recent study argues for simulations, labs and games to practice attack and defense safely, thus bridging theory and practice for hands-on, student-centered learning in cybersecurity education (Lazarov et al., 2025). As a result, learners can feel disengaged when content is static and more theoretical, limiting interactive problem-solving and learning experiences. For example, incorporating practical exercises, cyber-ranges, or gamified challenges has been shown to boost student motivation and interest in cybersecurity topics (Lazarov et al., 2025). In summary, by making learning more practical and adaptive, educators can address the main challenges of complexity and engagement.

## 2.2 AI in cybersecurity education

In terms of applying AI as a tutor tool to enhance learning in this field, researchers have begun exploring intelligent tutoring systems and AI-driven content tailored to cybersecurity topics in recent years. For example, researchers have used generative AI to create cybersecurity scenarios automatically, such as simulating network attacks for students to generate personalized vulnerability analysis tasks (Mukherjee et al., 2025). These AI integrations are designed to provide on-demand support and to scale practical training. Because cybersecurity is hands-on and scenario-driven, AI can generate diverse practice scenarios, for instance, varying the difficulty level of malware attacks in each exercise as students progress through their learning journey (Mukherjee et al., 2025). One notable implementation is a multi-agent tutoring system that orchestrates several specialized GPT-4-based agents, each expert in a subdomain (e.g., cryptography, network security), to assist graduate cybersecurity students through complex, multidisciplinary problems (Mukherjee et al., 2025). Because cybersecurity is hands-on and scenario-driven, AI can help generate diverse practice scenarios, for instance, varying the parameters of a malware outbreak exercise for students. Such variety is difficult for a human instructor to create repeatedly, but AI can produce numerous realistic variants (Mukherjee et al., 2025). The same study suggests that AI can be a double-edged sword despite its effectiveness in creating realistic training and exercises. For instance, pilot implementations of AI-driven cyber labs show that students can engage more deeply when scenarios are interactive and tailored (Mukherjee et al., 2025). At the same time, researchers emphasize on developing AI systems that are flexible and secure. The system should be able to handle a range of cybersecurity content, ensure the privacy of student data and allow lecturers to update content as threats evolve (Mukherjee et al., 2025). Additionally, research in learning analytics emphasizes that monitoring such behavioral indicators can help identify under-performing students much earlier than waiting for the end of their exams (Bergdahl et al., 2024). Therefore, the AI-CES can provide the human instructor with a starting point for intervention when some students have requested maximum hints on three consecutive phishing email tasks. In summary, such approaches aim to increase trust in AI among both lecturers and students, keeping lecturers firmly in the loop to oversee AI's contributions.

## 2.3 Generative AI solutions in education

The use of AI in educational tools and AI-assisted delivery is becoming increasingly common in the learning and training sectors. Mello et al. (2023) discussed the rise of Generative AI (GenAI) in educational settings, noting its adoption by major platforms such as Coursera, Khan Academy, and Duolingo. These platforms have integrated GenAI tools to enable personalized learning, provide real-time assessments and feedback (Newar et al., 2023), enhance predictive analytics, and deploy virtual teaching assistants, which aligns well with the educational needs of cybersecurity programs. These systems enhance learning by offering personalized support and adaptive feedback, helping students progress at their own pace while deepening their understanding of complex subjects (Khan Academy, 2025). Similarly, Duolingo utilizes its birdbrain Long Short-Term Memory (LSTM) AI to tailor learning experiences according to individual user needs, further illustrating the potential of GenAI to create adaptive educational environments (Bicknell et al., 2023). Currently, there are tools like "ChatGPT for Educators" and chatbots designed for content generation and tutoring (OpenAI, 2024). While these chatbots can be helpful, they mostly focus on isolated instructional activities or evaluation tasks. They may lack a comprehensive framework needed to scale effectively in larger, multidisciplinary educational contexts, especially within cybersecurity graduate programs (Yu and Guo, 2023). Recent study by Greco and Chianese (2024) highlighted the importance of human oversight when deploying AI-generated educational system. Their security assessment of an LLM-Generated e-learning application identified several vulnerabilities related to access control, content exposure, and authentication, demonstrating that AI-generated outputs must be carefully reviewed before being used in education environments. These findings reinforce the need for maintain human supervision when integration AI into cybersecurity education.

## 2.4 Enhancing student engagement

Research by Diakoumakos (2023) and Vykopal et al. (2022) emphasizes the importance of adaptive learning environments and AI-assisted gamified systems in cybersecurity education. These approaches not only facilitate personalized learning but also help students develop the necessary skills, thereby improving engagement and educational outcomes, which ultimately enhances student retention and meets industry needs. The literature indicates that while Generative AI (GenAI) shows great potential for transforming cybersecurity education by bridging the gap between theory and practice, there are still challenges in its systematic application and evaluation. Building on these insights, Mukherjee et al. (2025) proposed a platform intended to support scalability that includes methodologies for assessing the impact of GenAI on practical skill development and educational outcomes

in cybersecurity programs, ensuring that they effectively address both educational and industry demands. There are still challenges that require further exploration, particularly in validating the conceptual framework and assessing the real-world effectiveness of a Gen AI-driven Intelligent Tutoring System (Gen AI ITS) (Mukherjee et al., 2025). It is important to evaluate how effectively this system improves learning outcomes and the delivery of practical skills in cybersecurity. To do this, a combination of analysis and controlled experiments should be employed to evaluate how Gen AI influences students' learning over time. This paper address these challenges by proposing a system that integrates specific feedback mechanism and granular performance metrics to track students progress.

## 2.5 Performance tracking and feedback system

Unlike the proof-of-concept provided by Mukherjee et al. (2025), which lacked persistent user profiles, the proposed system incorporates a persistent tracking database to enable longitudinal analysis of student progress. Research in learning analytics emphasizes that monitoring such behavioral indicators can help identify at-risk students much earlier than waiting for exam results (Bergdahl et al., 2024). In addition, modern systems in higher education often use behavior metrics like logins, clicks and time on task as calculations for engagement (Bergdahl et al., 2024). Such behavioral metrics have demonstrated effectiveness, yet, it has its drawbacks: only quantitative metrics may fail to account for a student's emotional or cognitive involvement. To address this, more dashboard work is needed to bring in various dimensions, such as trying to gauge confidence or confusion from the hint patterns used. By synthesizing data, the dashboard can, for example, alert: "Student A has requested maximum hints on two consecutive SQL injection attacks." Such a nudge gives the human instructor a starting point for intervention. In the same study, it is stated that this measure improved learning outcomes, which the combined AI analytics and human oversight aims for by being responsive to signals in real time (Bergdahl et al., 2024). Moreover, another research indicates that implementing a dashboard can enhance engagement: when students know that their progress is monitored or can track it themselves, they tend to participate more regularly and effectively (Ramaswami et al., 2023). In summary, the instructions in the AI-CES can be tailored to meet each student's individual needs, ensuring that no one is overlooked. Additionally, the dashboard connects AI support with lecturers and offers feedback on student interactions. This approach may helps teachers keep students engaged and enhances their learning outcomes.

## 2.6 The educational platform development features

A key barrier to the widespread adoption of AI-powered ITSs today is the specialized programming and design skills required to create effective systems, as Calo and Maclellan (2024) highlighted. Moreover, while no-code or low-code authoring tools address programming limitations, Mukherjee et al. (2025) said that one cannot assume the educators possess the necessary design skills to create engaging and effective interfaces. While they only had a foundational understanding of chaining tools and agents as building blocks in modern application development, they implemented a practical application using low-code development tools like Streamlit and Revance AI (Mukherjee et al., 2025). Additionally, they used existing endpoints for OpenAI's APIs for models and chains like GPT-3.5 and GPT-4, and each was for the task at which they were most efficient. They developed two agents for supporting content ingestion and content generation, and the other agent was more conversational (Mukherjee et al., 2025). In addition, the CyATP solution, as an existing ITS solution, while innovative, utilized a singular, static dataset (CROND), thus limiting its adaptability and scalability (Beuran et al., 2022). Conversely, tools like ChatGPT for Educator demonstrate the effectiveness of dynamic personalization and adaptive content generation in educational contexts but have not yet been evaluated extensively in graduate-level cybersecurity education.

However, although their proposed system relies primarily on using prompt engineering techniques and enhancing personalization by leveraging interaction-based data gathering during the students' engagement with quizzes, the proposed system is designed to support dynamic content generation and continuous adaptability based on individual learner interactions, improving upon existing methods by enabling real-time differentiation and feedback. In summary, compared to prior systems such as CyATP, which rely on static datasets that limit adaptability and general tools such as ChatGPT for Educator, which lack educator oversight, raising concerns about accuracy, the proposed AI-CES offers a distinction by integrating a lecturer-on-the-loop architecture. Therefore, this ensures that while AI generates dynamic scenarios, all contents are validated by human instructors, limiting the gab between pedagogical accuracy and automated scalability. Unlike prior GenAI tutoring approaches that either focus on direct student-facing assistance or depend on relatively fixed content structures, AI-CES ensures lecturer approval before release, combines editable AI-generated cybersecurity simulations with role-based dashboards, and captures learner interaction data for instructor review. The contribution of this study is, therefore, not a claim of proven learning effectiveness, but the design and preliminary evaluation of a lecturer-supervised architecture for practice-oriented cybersecurity education.

# 3 Methodology

This study adopted a design-and-development approach to create and conduct an initial evaluation of the AI-CES prototype. The methodological focus was on two purposes: first, to design and implement a lecturer-supervised AI-assisted platform for scenario-based cybersecurity learning; and second, to obtain preliminary evidence on its feasibility, usability, perceived content quality, and perceived educational value in a higher education setting. The study was, therefore, positioned as an exploratory pilot rather than as a controlled effectiveness trial. The methodology was structured

to address the study's research questions in a way that matched the scope of the prototype. The design and implementation phases addressed the technical and pedagogical requirements of a lecturer-in-the-loop cybersecurity learning platform, while the evaluation phase examined how lecturers and students perceived the usefulness and usability of the system during initial deployment.

## 3.1 System design and implementation

The AI-CES prototype was developed as a full-stack web application consisting of four main stages: user interface development, AI services integration, user management, and hosting infrastructure. This architecture was selected to support flexibility, role-based interaction, rapid prototyping, and cloud-based access for both lecturers and students.

### 3.1.1 Stage 1: User interface development

The front end was developed using React.js as a single-page application. Separate interfaces were designed for lecturers and students in order to support their different tasks within the platform. Lecturer-facing pages supported simulation creation, preview, editing, approval, and access to student activity data. Student-facing pages supported topic selection, simulation gameplay, hint use, answer submission, and progress viewing. Navigation between views was handled through React Router. Local state management was handled with React's "useState," while the Context API was used for cross-component state, such as storing the authenticated user role across the application.

The back end was implemented using Node.js and Express.js. Its main role was to receive requests from the front end, communicate with third-party AI services, process simulation data, and store user activity logs. Each simulation round was structured to include a scenario prompt, multiple response options, a timer, and optional hints and feedback. Backend modules were separated by function. For example, one module handled OpenAI API requests for simulation generation, another managed voice synthesis requests, and another handled the logging of student interactions such as simulation completion and answer submission.

### 3.1.2 Stage 2: AI services integration

OpenAI's GPT-4 model was integrated as the core text-generation service within AI-CES. Lecturers entered a prompt describing a cybersecurity topic or scenario, and the system used this input to generate a structured multi-round simulation in JSON format. This output was then processed by the back end and stored as a draft simulation for lecturer review. To support multi-modal delivery, ElevenLabs was integrated for voice narration. Text from each simulation round was sent to the voice service and returned as an audio file. These files were then stored in cloud storage and linked to the relevant simulation rounds, allowing students to access narrated versions of the content when needed.

### 3.1.3 Stage 3: User management and data storage

Firebase Authentication was used to manage user accounts and role-based access. When users logged in, the system assigned them a role as either lecturer or student and used this role information to control access to relevant views and functions. Authentication tokens were also used by the back end to verify requests and protect lecturer-only endpoints. Cloud Firestore was used as the main database for persistent storage. It stored user profile information, simulation content, approval status, and student interaction logs. The database also enabled the capture of activity data such as simulation start and completion, answer submission, hint use, and time spent on each round. This supported the lecturer dashboard and enabled real-time updating of learner activity data.

### 3.1.4 Stage 4: Hosting and deployment

During early development, the system was tested locally. For wider accessibility during the pilot evaluation, the back end was deployed on Replit, while the front end was hosted using Firebase Hosting. This cloud-based deployment made the prototype accessible across multiple devices and locations. Although such deployment improved convenience for testing and demonstration, it also meant that the system depended on third-party service availability and internet connectivity.

## 3.2 Lecturer-in-the-loop workflow

A central design principle of AI-CES was lecturer supervision of AI-generated content. The platform was not designed to release AI-generated simulations directly to students without human review. Instead, lecturers first entered a topic prompt, after which the AI generated a draft simulation. Lecturers could then review, edit, regenerate, test, and approve the simulation before making it available to students. This workflow was intended to support pedagogical oversight and reduce the risk of inappropriate, unclear, or inaccurate AI-generated content. It also allowed lecturers to align generated scenarios with course terminology, expected difficulty level, and intended learning focus. Once approved, simulations became visible to students through the student dashboard. During student use, the system recorded interaction data, allowing lecturers to monitor engagement and performance through a tracking interface.

## 3.3 Research design for the pilot evaluation

The evaluation used an exploratory, cross-sectional design based on post-use feedback from intended users of the platform. The purpose of this phase was not to measure objective learning gains or establish causal impact, but to examine the feasibility of the prototype and gather preliminary evidence on how it was perceived by lecturers and students. Two role-specific digital surveys were used for data collection. The surveys were designed to reflect the different ways lecturers and students interacted with the system

while preserving common attention to usability, interaction quality, and perceived educational value. This design was chosen because the platform involved two different user groups with distinct responsibilities: lecturers as content creators and reviewers, and students as learners interacting with approved simulations.

## 3.4 Participants and setting

The pilot evaluation involved 30 participants recruited through convenience sampling from a computer science department, specifically from the cybersecurity area. The participant group consisted of 10 lecturers and 20 students. The lecturer group comprised faculty members with experience relevant to cybersecurity teaching, while the student group comprised undergraduate students studying cybersecurity-related content. The pilot session took place in Room 207 of the Harrison Building at the University of Exeter. The study was scheduled in a single session based on participant availability. Because the sample was drawn from one department and one institutional setting, the findings should be interpreted as preliminary and context-specific rather than generalizable across institutions or learner populations.

## 3.5 Evaluation instruments

Data were collected using two custom-designed surveys, one for lecturers and one for students. The surveys were informed by common dimensions used in educational technology and usability evaluation, including usability, engagement, perceived usefulness, interaction experience, content quality, and trust-related issues. Each survey included demographic items, Likert-scale questions, and open-ended response items.

The lecturer survey was organized around the following areas. Pedagogical alignment and curriculum relevance examined whether the simulations were perceived as suitable for cybersecurity teaching and aligned with intended learning outcomes. Classroom integration explored whether the platform could be incorporated into teaching practice and used to support topic-specific instruction. Perceived student learning support focused on whether lecturers believed the system could support student engagement and understanding. Technical usability and support examined ease of use, clarity of interface, and smoothness of operation. AI-generated content quality focused on the perceived relevance, clarity, and usefulness of AI-generated text and voice content. AI adaptivity and personalization examined the perceived value of hints, feedback, and adaptation within the platform. AI trust and ethics explored concerns relating to fairness, privacy, reliability, and responsible use.

The student survey was organized around the following areas. System usability examined ease of navigation and use of the student interface. Engagement and immersion explored how enjoyable and interactive the simulation experience felt. Feedback and adaptive support focused on the usefulness and clarity of hints, feedback messages, and pacing. Perceived learning support examined whether students felt the simulations helped them understand cybersecurity concepts and feel more confident in dealing with cybersecurity problems. Text, voice, adaptivity, trust, and fairness focused on perceptions of AI-generated content and the broader experience of interacting with the system. Because this was an early-stage pilot study, the surveys were used primarily to gather structured perception data rather than to produce validated psychometric measures of learning impact.

## 3.6 Procedure

At the start of the pilot session, participants were given an overview of the AI-CES platform and its main functions. This introduction was intended to ensure that both lecturers and students understood how to navigate the system and use the features relevant to their roles. The demonstration was tailored to each group. Lecturers were shown how to generate, review, edit, and approve simulations, as well as how to access performance tracking features. Students were shown how to select an approved simulation, interact with the rounds, use hints and audio support, and view their own progress. After that, participants were provided with access to the platform and asked to interact with it according to their designated role. Lecturers explored the simulation generation and review workflow, while students completed approved simulations through the student interface. Following this hands-on interaction, participants completed their respective digital survey. Survey responses were collected electronically using Google Forms and exported for analysis.

## 3.7 Data analysis

Quantitative and qualitative data were analyzed separately and then interpreted together to provide an overall account of the pilot evaluation. The quantitative survey data were organized in Google Sheets and analyzed using descriptive statistics. Frequencies, percentages, mean scores, and standard deviations were calculated to summarize responses across the main survey dimensions. Given the exploratory nature of the study, the modest sample size, and the absence of a comparison group, the analysis was intentionally descriptive rather than inferential. The quantitative results were, therefore, used to identify broad response patterns rather than to make causal claims about educational effectiveness. The open-ended responses were analyzed using a simple thematic analysis approach suitable for an exploratory pilot study. Responses were first reviewed in full to gain familiarity with the data. They were then grouped according to recurring patterns relating to usability, engagement, content clarity, feedback, trust, and suggestions for improvement. These recurring themes were subsequently summarized to capture the main issues raised by participants across both user groups. The qualitative analysis was used to add contextual depth to the survey ratings and to identify practical design implications for future versions of the platform.

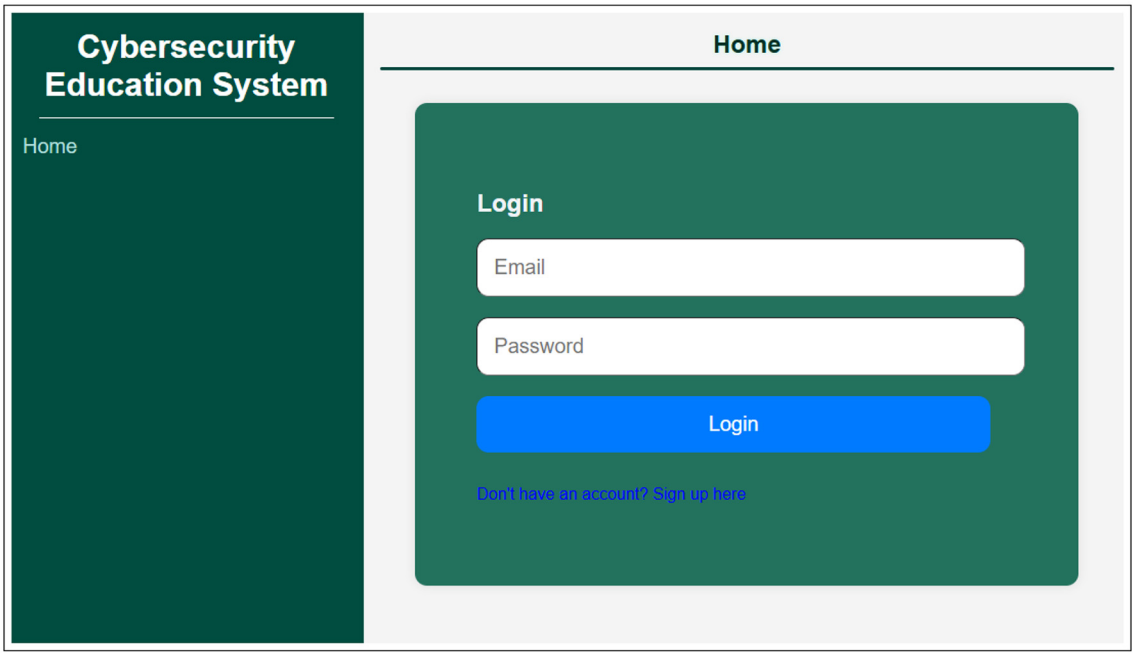



FIGURE 1
Users login interface of the AI-CES.

### 3.8 Scope of the evaluation

The evaluation was designed to examine prototype functionality and user perceptions following initial use of the system. It was not designed to determine whether AI-CES improves learning outcomes relative to other teaching ways, nor did it include pre and post-test measurement, a control condition, or long-term tracking of retention. In fact, the findings provide preliminary evidence of feasibility and acceptability, while also identifying areas for refinement and more rigorous future evaluation.

## 4 Results

This section presents the main findings from the development testing and pilot evaluation of the AI-CES prototype. The results are organized into two broad areas: the functional performance of the AI-CES platform and users' perceptions of its usability and perceived educational value. In keeping with the exploratory nature of the study, the findings are presented as evidence of feasibility, functionality, and user acceptability rather than proof of learning effectiveness.

### 4.1 The functionality of the AI-CES platform

Testing showed that the main components of the AI-CES prototype operated as intended during development and pilot use. The frontend displayed AI-generated content, accepted user input, and transmitted interaction data to the backend. The backend processed these requests, updated the database, and refreshed the lecturer dashboard through Firestore's real-time update mechanism. Core functions including role-based login, AI-assisted simulation generation, lecturer review and approval, and student activity tracking were all tested successfully.

#### 4.1.1 The testing and evaluation of the user interfaces functionalities

To evaluate interface functionality, role-based login and access control were tested using dummy accounts. A user with lecturer credentials was able to access lecturer dashboard pages, while a student user was directed to the student homepage. It was also verified that a student account could not access lecturer-only endpoints through a direct URL request, as the backend validated the Firebase token and returned an unauthorized error when access conditions were not met. Figure 1 shows the role-based login interface which allows users to login the platform according to their roles.

The AI-CES prototype integrated AI-assisted simulation generation, role-based access control, and learner activity tracking within two main dashboards. Figure 2 shows the lecturer dashboard, which provided access to functions such as "Add a new Simulation," enabling lecturers to enter a scenario description and generate a draft simulation topic. Once generated, lecturers could review, edit, regenerate, and approve the simulation before releasing it to students. The "Tracking Performance" tool provided a real-time view of student activity and progress. Overall, this lecturer dashboard shows the content management and student performance monitoring features.

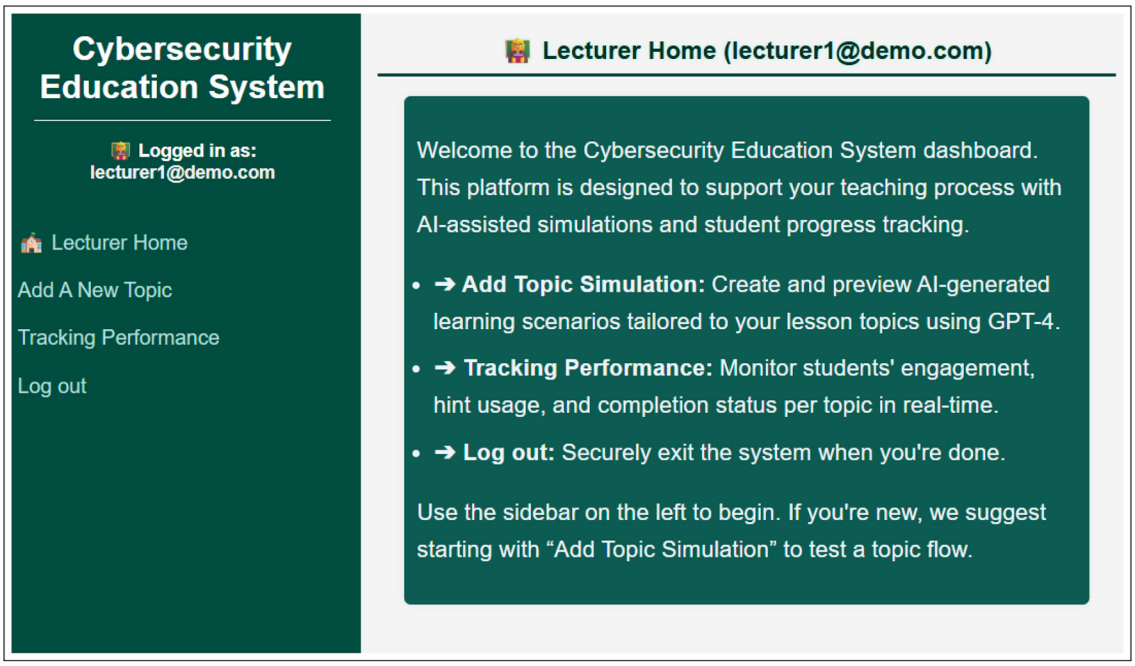



FIGURE 2
Lecturer dashboard interface of the AI-CES.

Figure 3 shows the student dashboard, which served as the main interface for interacting with approved simulations created under lecturer supervision. Students could select topics through the "Start New Simulation" option, review previous activity through the "View Progress" tab, and complete multi-round scenario tasks using a one-minute timer, audio playback, hint support, and answer submission. It also displayed newly released topics associated with a lecturer. In general, this student dashboard provide access to cybersecurity simulation activities and progress tracking.

### 4.1.2 The testing and evaluation of the AI content

GPT-4 was used to generate simulation content for students based on lecturer prompts. It produced structured multi-round simulations, including scenario descriptions, response options, and round-level feedback. This design supported the creation of varied cybersecurity scenarios and enabled immediate feedback following each student response. In functional testing, clearly written prompts generally resulted in coherent multi-round simulations. For example, a prompt on a ransomware attack produced a three-round scenario covering vulnerability scanning, exploitation, and post-attack response, with plausible options across the sequence. Figures 4, 5 present examples of the first and second rounds in an AI-generated ransomware simulation. These figures illustrate how students received immediate system feedback after submitting a response. They also show that different rounds addressed different aspects of the same topic while remaining aligned with the core subject focus defined by the lecturer.

Overall, the generated content was generally usable for the prototype context, although occasional issues were observed. Such as overly complex wording, discipline-specific jargon, or phrasing that required simplification before student release. The lecturer approval stage was, therefore, an important part of the workflow, allowing content to be edited before publication, as illustrated in Figure 6. This figure illustrates the lecture approval and editing interface for reviewing AI-generated simulation content.

Most of the technical content generated by the model appeared sufficiently accurate for prototype use, suggesting that GPT-4 can support the drafting of domain-specific educational scenarios when lecturer oversight is retained. The narration files generated through ElevenLabs were also functional within the prototype. These audio files were pre-loaded at the start of simulations and loaded quickly on a typical broadband connection. On slower mobile networks, there was occasionally a short delay of around two seconds before playback became available, which remained acceptable for a supplementary accessibility feature. In rare cases, voice generation failed, in which case the system hid the play button for that round and logged the error for review.

### 4.1.3 The testing and evaluation of database functions

Testing of the Cloud Firestore database showed that the backend was able to capture and store detailed learner interaction data. Logged events included simulation activity and other interaction records associated with student use. The system was also accessed from different locations and devices, including laptops and smartphones, suggesting that the cloud-based design supported distributed use during the pilot stage. Despite this dependency of the third-party cloud services, the prototype successfully logged and stored user activity throughout platform use.

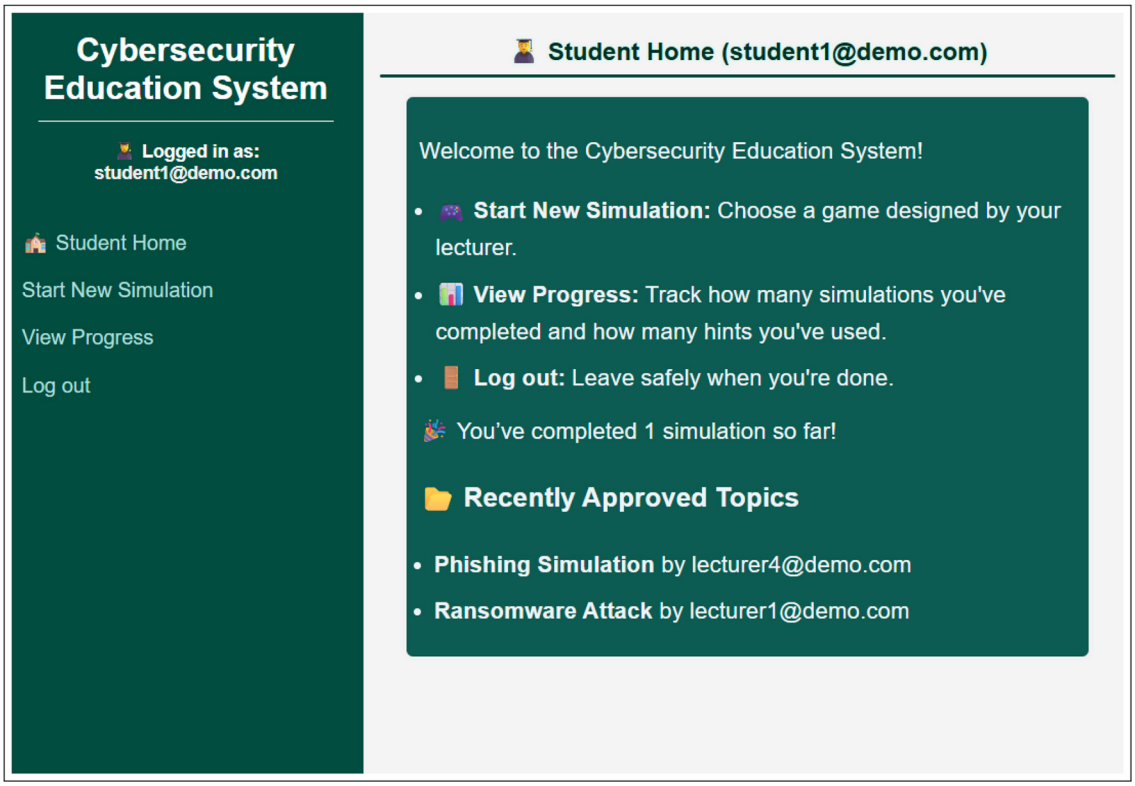



FIGURE 3
Student dashboard interface of the AI-CES.

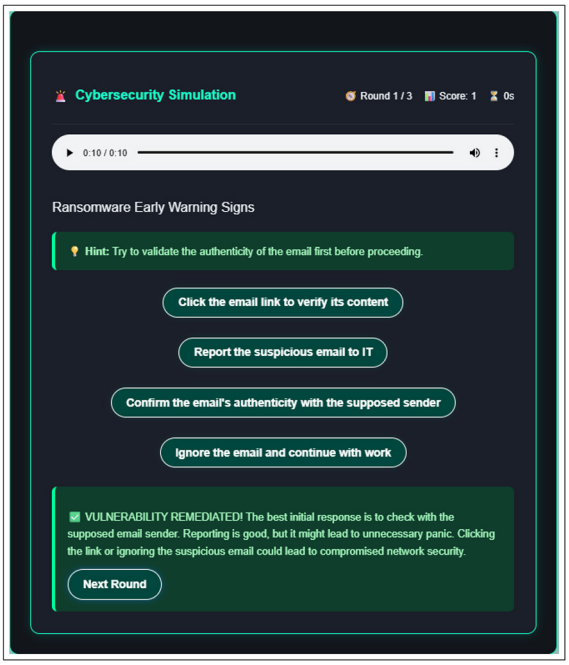



FIGURE 4
First round of an AI-generated ransomware scenario.

## 4.2 The usability of AI-CES platform

The usability evaluation of AI-CES was conducted across several cybersecurity topics, including ICMP Flood Attacks, Ransomware, Phishing, and Social Engineering. Usage data such as completion records and scores were generated through system interaction, while perception data were collected through two structured digital surveys administered to lecturers and students. Survey responses were collected through separate Google Forms questionnaires. Quantitative data from the Likert-scale items, using a scale from 1 = Strongly Disagree to 5 = Strongly Agree, were analyzed using descriptive statistics. Open-ended responses were analyzed using a simple thematic approach to identify recurring patterns in participant feedback. For the descriptive usability summary, responses were aggregated across the usability-related survey items. On this basis, lecturer responses produced a mean usability score of approximately 4.21 (SD = 1.18), indicating a broadly positive view of system usability and functionality. Most lecturer responses were concentrated in the higher agreement categories, particularly ratings of four and five, suggesting that the platform was generally perceived as intuitive, useful, and supportive of teaching-related tasks. Student responses also showed a positive overall pattern, with a mean usability score of approximately 4.11 (SD = 1.04). Most student ratings were similarly clustered around the agreement categories, indicating that the platform was generally

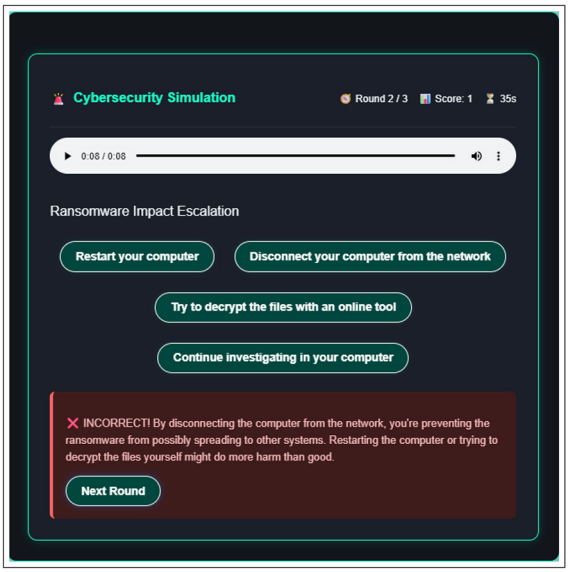



FIGURE 5
Second round of an AI-generated ransomware scenario.

perceived as easy to use, engaging, and helpful for interacting with cybersecurity scenarios. Figures 7, 8 illustrate the distribution of lecturer and student usability ratings for the AI-CES platform.

Although some variability was evident across individual items, the overall pattern suggested that both user groups responded favorably to the platform in this pilot setting. In addition to the quantitative analysis, open-ended responses and immediate participant comments were reviewed using a simple thematic analysis approach. Moreover, the qualitative feedback provided additional insight into how users experienced the platform during initial use. Nevertheless, these results should not be interpreted as evidence of improved learning outcomes.

However, on the one hand, lecturer feedback highlighted two recurring themes: ease of content authoring and the value of learner tracking. Several lecturers reported that it was easier to create a simulation prompt than to design a complete practical activity manually. One lecturer remarked that writing a scenario prompt was "much easier than designing a full lab by hand." Another commented that "the AI-generated questions provided a useful starting point, and I only needed to make small adjustment or refinement before approve them to students" for the scenario requested. At the same time, some lecturers reported making small edits to improve wording or align terminology with their own course content, which reinforced the value of the editable preview stage. A lecturer commented that, "The approval step gave me confidence in the quality of the content because I could review, adjust, refine a topic before it was released." Feedback on the tracking dashboard was also positive, particularly in relation to identifying students who might need support. For example, a lecturer remarked that "The design of the interface of the platform helped me identify students who were struggling with specific topics much more quickly than traditional assessment methods." Similarly, another lecturer noted that "If a student consistently takes maximum time or uses hints, I know to reach out." Despite that, some suggested improvements to the interface or the dashboard analytics, noting that the event log could be made more easier to interpret and provide a clearer view of student performance because some information appeared overly aggregated in its current form. However, some lecturers explained that "Overall, the combination of AI-generated output and their oversight created a good balance between efficiency and academic quality." These feedback supports the perceived usefulness of the AI-CES for both student monitoring and content creation.

On the other hand, student feedback highlighted the value of immediate feedback, interactivity, and multi-modal support. Many students commented positively on receiving round-by-round feedback, which helped them understand why an answer was correct or incorrect. One student described the system as a "mini-game that teaches you along the way." Students also reported that the simulation format made the learning experience feel more interactive than static content. However, some comments suggested that the hint feature might be used too quickly when students felt unsure, indicating a possible need to refine how hints are timed or presented. One student commented, "I saw a hint as soon as I was unsure, maybe too late." Students generally viewed the audio narration as a useful addition, although some indicated that they would be more likely to use it for longer passages of text.

Taken together, the usability findings suggest that the AI-CES prototype was well received by both lecturers and students as an exploratory lecturer-supervised platform for scenario-based cybersecurity learning. Table 1 shows a comparison summary of the main themes identified from students and lecturers feedback, highlighting similarities and differences in perceptions of the AI-CES platform. These findings indicate that the prototype is feasible and acceptable in an initial pilot context, while also pointing to areas for refinement in future iterations.

# 5 Discussion, limitations, work future and conclusion

## 5.1 Discussion

The findings from this pilot study suggest that AI-CES is a feasible and promising lecturer-supervised platform for scenario-based cybersecurity education. Rather than demonstrating confirmed educational effectiveness, the results provide preliminary evidence that the prototype can support interactive content delivery, lecturer oversight of AI-generated materials, and student engagement with practice-oriented cybersecurity scenarios. This interpretation is consistent with recent literature showing that cybersecurity education benefits from more hands-on, adaptive, and practice-oriented learning environments, while also facing persistent challenges related to engagement, communication, and the translation of theory into practice (Crabb et al., 2024; Salem et al., 2024; Vykopal et al., 2022).

First, the platform demonstrated the practical value of a lecturer-in-the-loop design. Lecturers were able to generate, review, edit, and approve AI-generated cybersecurity scenarios before releasing them to students. This design is important because it preserves pedagogical oversight while still benefiting from the speed

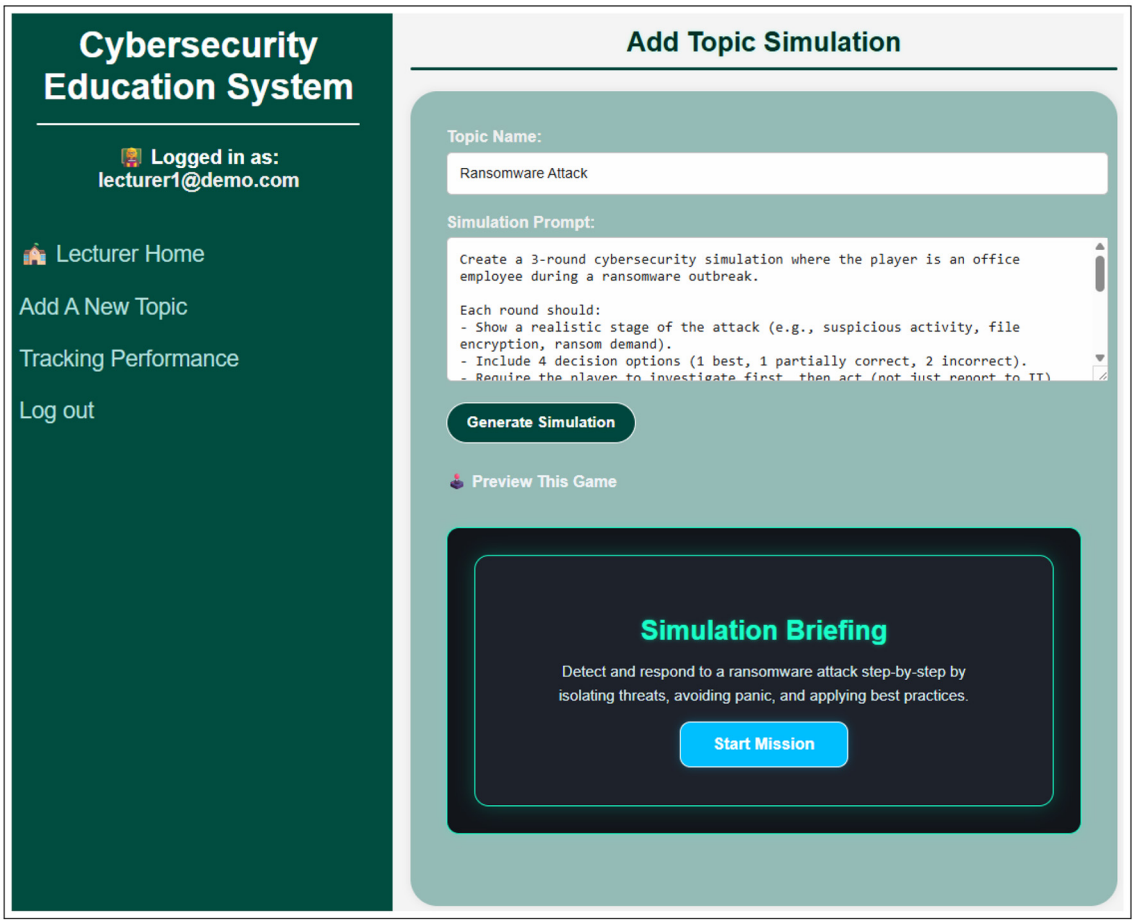



FIGURE 6
Lecturer review and editing interface for AI-generated scenario content.

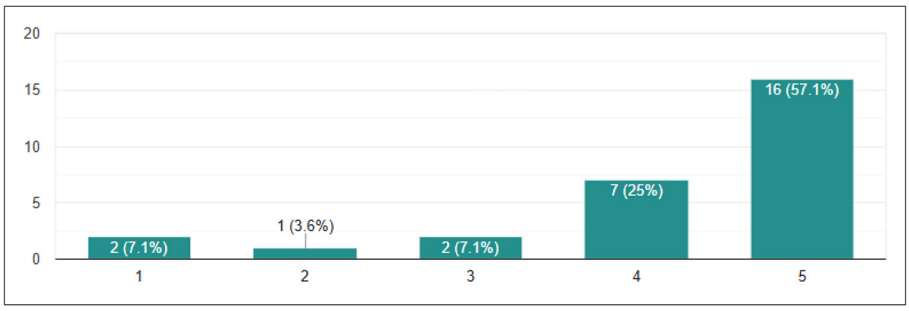



FIGURE 7
Distribution of lecturer response to the AI-CES usability evaluation survey.

and flexibility of generative AI. The platform, therefore, supported a collaborative workflow in which AI assisted with draft content generation, while lecturers retained responsibility for ensuring relevance, accuracy, and alignment with course expectations. In turn, students interacted only with content that had already passed through a human review process. This feature distinguishes the platform from fully automated student-facing systems and aligns with recent work arguing that generative AI can support educator-driven tutor authoring, but should do so in ways that keep educators actively involved in design and decision-making (Calo and Maclellan, 2024). It also complements emerging work in cybersecurity education that positions generative AI as a potentially scalable support mechanism, while still recognizing the need for careful evaluation and responsible deployment (Mukherjee et al., 2025).

Second, the study showed that the integration of GPT-4 and ElevenLabs was technically workable within the prototype. GPT-4 was able to generate coherent multi-round simulations from

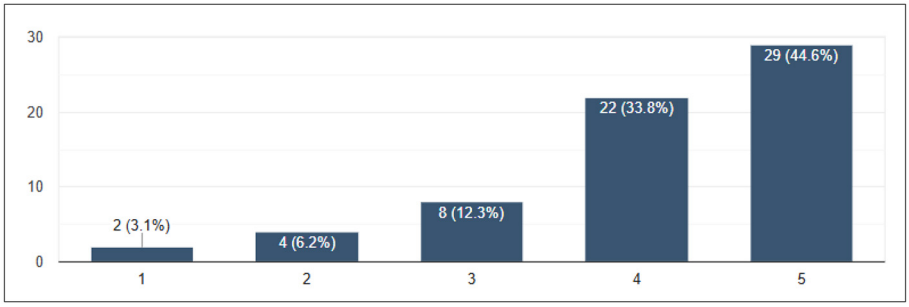



FIGURE 8
Distribution of student responses to the AI-CES usability evaluation survey.

TABLE 1 Summary comparison of usability evaluation results between lecturers and students.

| Usability and educational dimensions | Students | Lecturers |
|---|---|---|
| Quality of learning content | Good quality and relevance of cybersecurity scenarios | Good overall quality, although some AI-generated content required minor editing before release |
| Engagement and interactive | Strongly positive; several students described the experience as game-like and engaging | Moderately positive; lecturers appreciated the interactive learning approach |
| Ease of AI-CES usability | Favorable; most reported that it was easy to navigate and use | Generally favorable; they reported that content creation was easier than creating activities manually |
| Analytics and tracking | Not directly applicable as this feature only for lectures | A valuable for observing students performance, although some requested more aggregated views |
| Trust in AI voice narration | Generally Positive, although some preferred it mainly for longer content | Considered a useful supplementary feature |
| Quality of feedback | Highly valued for explaining correct and incorrect answers during the rounds | Useful for identifying learning difficulties and monitoring performance |

lecturer prompts, while ElevenLabs added an audio layer that some participants found useful for accessibility and engagement. At the same time, the results also showed why human oversight remains necessary. Although the generated content was generally usable, some outputs contained overly complex wording or jargon that needed lecturer refinement before release. This reinforces a wider point in the literature that even advanced models such as GPT-4 can produce unreliable or inconsistent outputs in educational settings, particularly when tasks require sustained coherence, accurate reasoning, or trustworthy judgment (Gandolfi, 2025). Similarly, this aligns with observation of Greco and Chianese (2024) who illustrated that AI-generated educational systems may provide security and governance challenges if deployed without sufficient human review. Therefore, the platform's value of the AI-CES lies not simply in automation, but in combining AI-assisted generation with educator review and control. However, while the AI-CES prototype uses human approval as a safeguard for educational output and content, future deployment should further strengthen security controls, institutional oversight and data governance mechanism to ensure privacy, reliability and long-term sustainability.

Third, the tracking and dashboard features appeared to support a more informed teaching process. Student interaction data, including completion records, hint use, and timing information, were captured and made visible to lecturers through the platform. Lecturer comments suggested that this feature could help identify students who may need additional support and could make learner activity more visible than in static or one-directional teaching formats. This is broadly in line with learning analytics research showing that dashboards can improve visibility into learner engagement and support more timely intervention (Ramaswami et al., 2023). Students also responded positively to the immediate round-by-round feedback, suggesting that the system supported a more interactive and responsive learning experience.

However, these findings should be interpreted carefully because the present study did not include objective measures of learning gain, so the results support claims about perceived usefulness and interaction quality rather than verified improvement in performance or knowledge. The pilot also highlighted a number of practical strengths of the prototype. The storytelling structure of the simulations appeared to make activities feel more interactive than conventional static quizzes. The web-based design supported access across multiple devices, including laptops and smartphones, which may reduce some of the access barriers associated with more specialized cybersecurity training systems. In addition, the ability to generate new scenarios relatively quickly may be especially useful in cybersecurity education, where topics and threat contexts change rapidly (Crabb et al., 2024; Salem et al., 2024). These features suggest that AI-CES has a practical promise as a flexible supplementary teaching tool.

## 5.2 Limitations

Despite those promising findings, this study has several important limitations must be acknowledged. Firstly, the evaluation was conducted with a relatively small participant group from a single computer science department, which limits the generalizability of the findings to broader educational context. Therefore, the results should be interpreted as an initial feasibility evaluation of the AI-CES prototype rather than a large-scale validation of educational effectiveness. Secondly, the study relied mainly on survey-based usability and experience feedback without incorporating a control group or longitudinal pre and post-testing. Although the findings refer positive engagement and perceived learning support, more empirical validation is still needed. Thirdly, the system's dependence on third-party cloud such as OpenAI, ElevenLabs, Replit and Firebase. While these platforms significantly accelerated the implementation and testing stages of the prototype, long-term scalability may raises concerns regarding operational cost, dependency on external providers, and service availability. Future implementations may explore institutionally hosted or even self-managed alternative to improve sustainability, flexibility, and deployment control. Fourthly, to support secure access and user management, the AI-CES prototype utilized Firebase authentication and Firestore access controls to restrict system access to authorized users only. Educational interaction data collected during the experiment were limited to learning-related activities such as simulation progress, responses, hint usage, and completion records. No sensitive personal information beyond the educational requirements of the prototype was intentionally collected. However, future work versions should incorporate stronger institutional governance policies, clearer consent procedures, and enhanced privacy management mechanism to ensure compliance with educational data protection and ethical standards. Finally, some student comments suggested that the hint system may need refinement to discourage over-reliance or passive response behavior. Although the lecturer-in-the-loop workflow reduced risks associated with hallucination or inappropriate AI output, it did not eliminate wider concerns relating to bias, trust, or the pedagogical appropriateness of generative models (Gandolfi, 2025; Mukherjee et al., 2025). However, the study shows that the AI-CES can function as intended and that both lecturers and students responded positively to its initial use.

## 5.3 Future work

Future research should extend this work through larger-scale empirical studies involving participants from several institutions and cybersecurity programs. Moreover, future evaluations should incorporate control groups, longitudinal pre-test and post-test measures, and further rigorous statistical analysis to better measure the educational outcomes of the AI-CES over time. In addition, future work should also investigate secure deployment architecture, institutionally hosted AI models or managed alternatives to third-party AI services to improved privacy governance and reduce operational risks. Furthermore, adaptive learning analytics and broader cybersecurity simulation scenarios may enhance reliability, scalability, and long-term education adoption.

## 5.4 Conclusion

This study presented the design and preliminary evaluation of AI-CES, an AI-assisted interactive platform developed to support cybersecurity education through lecturer-supervised scenario-based simulations. The platform combined AI-generated draft content, lecturer review and approval, student-facing interactive tasks, and performance tracking within a single web-based environment. The pilot findings suggest that the system is feasible, functional, and positively received by its intended users in an exploratory context. The main contribution of the study lies in demonstrating a practical lecturer-in-the-loop architecture for the use of generative AI in cybersecurity education. Rather than replacing educators, the platform was designed to support them in creating and managing interactive learning activities while maintaining control over the quality and appropriateness of the content released to students. In this respect, AI-CES offers a useful proof of concept for how generative AI may be incorporated into practice-oriented digital education while retaining human oversight (Calo and Maclellan, 2024; Mukherjee et al., 2025). At the same time, the study does not establish that AI-CES improves learning outcomes or teaching effectiveness. The evidence presented here should instead be understood as preliminary evidence of feasibility and user acceptability. Future work will, therefore, focus on more rigorous evaluation through larger samples, controlled comparisons, multi-institutional deployment, and objective measures of learner performance and retention. With these next steps, the AI-CES approach may contribute to the development of more interactive, adaptable, and pedagogically grounded cybersecurity education.

# Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author/s.

# Ethics statement

The studies involving humans were approved by Engineering Mathematics and Physical Sciences Ethics Committee, University of Exeter. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

# Author contributions

YA: Methodology, Data curation, Formal analysis, Visualization, Software, Writing – review & editing,

Conceptualization, Writing – original draft. SO: Conceptualization, Investigation, Supervision, Writing – review & editing.

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## Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

## Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. This paper was written and reviewed by the named human authors. During the revision process, Generative AI tools were used to assist with language polishing, formatting, and LaTeX structuring. All intellectual contributions, including study selection, analysis, interpretation, and final writing, reflect my own work. The use of generative AI tools did not extend to content generation or literature synthesis.

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